From Aggregate Accuracy to Branch Reliability: A Capability-First Training Paradigm for Mixture-of-Experts Physics-Informed Neural Networks

Xiang Li[a], Liping Huang[a,*]

[a]*Henan University of Economics and Law, Zhengzhou, China*

Abstract

Soft mixture-of-experts physics-informed neural networks (MoE–PINNs) may produce accurate aggregate solutions while containing expert branches that are unreliable as standalone numerical solvers. We formulate a branch-reliability audit that separates aggregate validity, complete-branch capability, gate–capability alignment, and robustness to routing interventions. The audit identifies a self-reinforcing failure mechanism in end-to-end co-adaptive training: early gate preferences attenuate weak-branch gradients, while simultaneous expert updates continually alter the capability terrain presented to the gate. We therefore propose capability-first staged training (CFST), which forms numerical capability in complete branches before routing competition, then freezes the branches while learning the gate. In ten paired random-seed experiments on the two-dimensional viscous Burgers equation, co-adaptation attains a lower soft-mixture relative L_2 error (0.1600 ± 0.0379 versus 0.2108 ± 0.0252 for CFST), but its worst complete-branch error reaches 21.51 ± 8.34, approximately 64.5 times the CFST value of 0.3336 ± 0.0426. An exact error decomposition shows negative cross-branch energy in every co-adaptive run, cancelling $17.2\% \pm 10.1\%$ of diagonal error energy, whereas CFST exhibits no such cancellation. Under Top-1 routing, gate flattening, and deletion of the highest-load branch, co-adaptive error ratios are 1.38, 2.90, and 1.52, compared with 1.06, 0.99, and 1.09 for CFST. Allen–Cahn, KdV, and parameter-matched APINN controls support the same aggregate–branch distinction. The results establish an auditable qualification framework and a

*Corresponding author

Email address: hlp369369@163.com (Liping Huang)

capability-before-routing training paradigm, with CFST as an effective and computationally economical realization.

Keywords: Physics-informed neural networks, Mixture of experts, Branch-reliability audit, Capability-first training, Routing robustness, Domain decomposition

1. Introduction

Physics-informed neural networks (PINNs) encode partial differential equation residuals, initial and boundary conditions, and observational data into differentiable loss functions, and have become a basic paradigm in scientific machine learning [1, 2, 3]. However, when a single network is used for multiscale or multiphysics problems or discontinuous solutions, it is often hindered by gradient imbalance, spectral bias, and complex loss landscapes [4, 5, 6, 7]. Domain-decomposition strategies have emerged to alleviate this bottleneck: cPINNs, XPINNs, and FBPINNs prescribe geometric subdomains and interface coupling [11, 12, 13, 14], whereas APINN introduces a trainable partition-of-unity gate to realize a flexible soft domain decomposition [15].

Soft routing creates a distinction that aggregate benchmarks do not capture: a mixture may be accurate even when its expert decomposition is not reliable. The general mixture-of-experts (MoE) literature typically treats mixture prediction error and load balancing as core indicators of model quality [21, 22, 23, 24]. Neither quantity certifies that each complete branch is a usable numerical solver, that the gate selects branches according to capability, or that the solution survives a change in routing. These properties matter in scientific computing because soft routing may be hardened into Top-1 selection, branches may be removed to reduce cost, and gate weights may drift or be recalibrated after deployment. We therefore treat aggregate accuracy as a necessary but incomplete qualification criterion and ask a different question: does the learned expert decomposition remain numerically valid when its routing mechanism is interrogated or altered?

Figure 1 summarizes the co-adaptation trap and the staged optimization strategy developed below. We address two connected questions: how to evaluate whether a learned expert division is effective, and how to form an effective, routing-robust division at low computational cost. The branch-reliability audit separates aggregate validity, complete-branch capability,

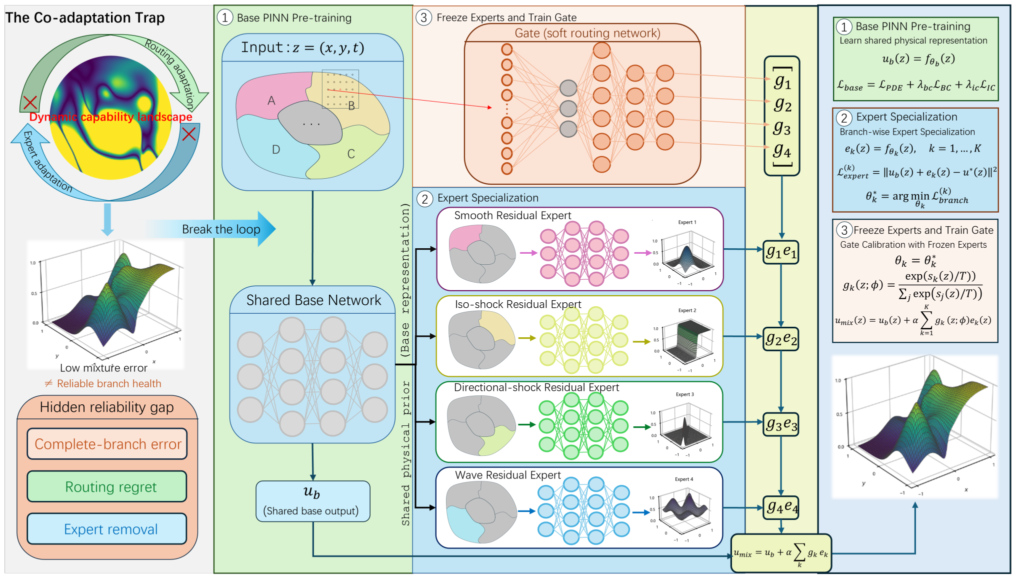

Figure 1: Overview of the co-adaptation trap in MoE–PINNs and the proposed staged optimization strategy. Soft gate–expert joint adaptation creates a moving capability landscape, where low mixture error may hide unreliable expert branches. The proposed staged optimization first learns a shared physical representation, then specializes residual experts, and finally calibrates the gate with frozen experts, decoupling routing optimization from expert capability evolution.

gate–capability alignment, and routing-intervention robustness. It then exposes a self-reinforcing failure loop in co-adaptive training: an early gate preference weakens the learning signal received by some branches, and their resulting capability deficit further reduces the routing mass they receive; simultaneously updated experts also present the gate with a moving capability terrain. Weak capability and low routing mass thereby reinforce each other.

Capability-first staged training (CFST) breaks this loop by forming complete-branch capability before routing competition begins. It first trains complete branches independently under branch-level physics constraints and reference information, then freezes the resulting capability terrain and trains the gate to route among competent branches. The protocol reorganizes training order and supervision topology without enlarging the architecture, yielding an effective, low-cost intervention. More generally, any mechanism that interrupts gate-induced suppression of branch learning, stabilizes the capability terrain seen by the gate, or maintains a lower bound on branch capability may realize the same improvement principle; two-time-scale optimization, stop-gradient routing, and explicit capability constraints are candidate implementations. CFST is the controlled realization studied here, whereas capability-before-routing defines the broader training paradigm.

The contributions are fourfold:

1. We formulate the qualification problem "aggregate accuracy does not certify expert decomposition" and make it operational through a branch-reliability audit spanning aggregate validity, branch capability, gate–capability alignment, and routing interventions.
2. We identify a self-reinforcing branch-failure mechanism in co-adaptive MoE–PINN training: gate weights scale expert gradients while simultaneous expert updates move the capability terrain seen by the gate.
3. We derive a capability-before-routing principle from PDE-function-space non-uniqueness, an exact nonlinear Burgers residual decomposition, a constructive counterexample, an oracle-label stability result, and a conditional branch-reliability certificate. CFST is a fully specified and experimentally validated realization of this principle.
4. We close the audit–mechanism–treatment loop with paired information controls, ten-seed Burgers comparisons, three same-grid routing interventions, Allen–Cahn and KdV transfer tests, and parameter-matched APINN controls.

The resulting paper follows a qualification-gap–mechanism–principle–treatment–validation sequence. Aggregate accuracy establishes that the solver remains competitive; branch-capability and routing-intervention audits determine whether the learned decomposition is qualified for modular use; CFST tests whether the capability-before-routing paradigm repairs the diagnosed failure.

2. Related work and positioning

2.1. PINN optimization and domain decomposition

Adaptive weighting, sampling, and causal curricula address parts of the PINN optimization problem [8, 9, 10], but do not determine how several learned branches acquire a stable division of labor. Geometric decomposition fixes branch responsibility, whereas soft decomposition makes it an optimization outcome. APINN shows that gate initialization can materially change performance [15]. A flexible interface learning framework further shows how adaptive interfaces can facilitate collaboration among domain-decomposed

PINN subdomains [16]. Our question is complementary: whether an accurate aggregate solution is sufficient to certify the reliability of the learned expert decomposition. CFST acts at the level of training order by forming complete-branch capability before asking a learned gate to divide the domain.

Jointly optimizing several learned branches also introduces gradient conflicts inside a single physical objective. The gate seeks to separate expert outputs (Section 3.2), while each expert seeks to fit the physical residual; the two gradient directions can oppose each other. Multi-task learning has developed remedies such as gradient projection and gradient surgery [17], but those methods modify conflicting gradients after competition has begun. CFST instead prevents early routing competition from controlling which branches obtain the opportunity to become competent.

2.2. General MoEs and MoE–PINNs

Adaptive local experts and hierarchical MoEs established joint gate–expert learning [18, 19]. Positive theory recovers specialization under latent cluster structure, suitable initialization, and favorable optimization conditions [27]. Absence of explicit regime labels does not prove absence of latent identifiable structure; rather, the assumptions needed by that positive theory are not established for the continuous PDE objectives studied here. MoEs have been used for PINN ensembles, soft decomposition, and locally hard physical constraints [32, 15, 33]. A recent analysis for one-dimensional hyperbolic conservation laws bounds the aggregated MoE–PINN prediction [34]; it neither requires every branch to be healthy nor excludes substantial mass on a poor branch. Two recent preprints extend physics-informed expert decomposition to output-aware soft gating for free boundaries and residual-driven hard gating for unknown interfaces [35, 36].

Load-balancing regularization in sparse MoEs primarily prevents capacity overflow and expert starvation [21, 23], yet representation collapse and routing degradation can still occur when loads are nearly balanced [28]. Load balance therefore cannot substitute for capability formation. Table 1 positions CFST relative to aggregate-error theory, specialization recovery, and existing physics-informed expert decompositions.

Table 1: Claim-level positioning relative to the closest lines of work.

Work	Main assumptions or object	Conclusion relevant here
Ensemble ambiguity [20]	Fixed ensemble members and aggregate squared error	Diversity can lower aggregate error; branch identifiability and learned routing are not addressed.
Specialization recovery [27]	Latent clusters, favorable initialization and optimization	Positive recovery under structured assumptions; those assumptions are not established for the present PDE objectives.
APINN [15]	Trainable partition-of-unity gate for soft domain decomposition	Gate initialization changes performance; no branch-health identifiability criterion is given.
Flexible interface learning [16]	Adaptive interfaces for collaborative domain-decomposed PINNs	Learns effective subdomain collaboration; complete-branch reliability under routing interventions is not evaluated.
MoE–PINN bound [34]	One-dimensional hyperbolic setting and aggregate predictor	Bounds the mixture prediction, not complete-branch reliability or routing counterfactuals.
Recent physics-informed MoE preprints [35, 36]	Soft/hard gated free-boundary and unknown-interface decomposition	Focus on geometric decomposition and solution recovery, not routing-intervention reliability of complete soft-mixture branches.
This work	Soft PDE mixture with trainable complete branches and routing	A branch-reliability audit exposes the aggregate–branch qualification gap; theory yields a capability-before-routing principle, and CFST provides its tested, low-cost realization.

3. Why aggregate accuracy does not certify branch reliability

3.1. Residual soft MoE–PINN

For $z \in \mathcal{Q} = \Omega \times (0, T)$, consider

$$u_{\mathrm{mix}}(z) = u_{\mathrm{b}}(z) + \alpha \sum_{k=1}^{K} g_k(z; \Phi) e_k(z; \theta_k),$$
$$g_k = \frac{\exp(s_k/T_g)}{\sum_j \exp(s_j/T_g)}, \qquad \sum_k g_k = 1. \tag{1}$$

The complete kth branch is $u_k = u_{\mathrm{b}} + \alpha e_k$, and $\bar{e} = \sum_k g_k e_k$. In the main Burgers model, $\alpha = 0.35$ and the learned-router temperature is $T_g = 0.8$; the separate teacher-target sharpness is denoted $\beta = 8$ below. Let the physical objective $\mathcal{J}$ collect PDE, initial, and boundary losses. In a neighborhood where its first variation exists, write $D\mathcal{J}(u)[v] = \langle q, v \rangle$ as a dual pairing.

Proposition 1 (Gate–expert coupling)**.** *Whenever the required derivatives* *exist,*

$$\nabla_{\theta_k}\mathcal{J} = \alpha(D_{\theta_k}e_k)^*[g_k q], \qquad D_{s_k}\mathcal{J}[\xi] = \left\langle q, \frac{\alpha}{T_g} g_k(e_k - \bar{e})\xi \right\rangle. \tag{2}$$

If q admits a density, the second identity is $\delta\mathcal{J}/\delta s_k = (\alpha/T_g)g_k(e_k - \bar{e})q$.

Persistently small soft weight attenuates the expert gradient[28], so Top-1 load cannot replace the continuous weight diagnostic. If all e_k are identical, the gate-logit gradient vanishes, but the expert gradients generally do not. The correct statement is therefore that expert-output symmetry suppresses gate learning, not that the full coupled system is necessarily stationary. Section 5 reports the corresponding training trajectories after the experimental quantities have been defined.

3.2. Non-identifiability under a mixture-only objective

Consider $\mathcal{L} = \mathcal{F}(u_{\text{mix}}) + \mathcal{R}(g)$, where $\mathcal{R}$ is gate-only and no branch-level supervision is present.

Theorem 1 (Function-space non-uniqueness)**.** *Suppose that an open set $U \Subset \mathcal{Q}$ and branches $i \neq j$ satisfy $g_i, g_j \geq \gamma > 0$ on U. For any nonzero $h \in C_c^\infty(U)$, define*

$$u_i' = u_i + h, \qquad u_j' = u_j - \frac{g_i}{g_j}h, \tag{3}$$

with all other branches and the gate fixed. Then $\sum_k g_k u_k' = \sum_k g_k u_k$, so $\mathcal{F}$, $\mathcal{R}$, and all gate statistics are unchanged. Scaling $h = M\psi$ and taking $M \to \infty$ makes at least one branch error unbounded while preserving the mixture.

Proof. The weighted perturbation is $g_i h - g_j(g_i/g_j)h = 0$. Because h is compactly supported in the interior, it vanishes near the parabolic boundary and preserves any hard initial or boundary values satisfied by the original predictions. Scaling completes the argument. □

Remark 1. This is a function-space theorem. Finite neural networks need not represent every compensating perturbation exactly. The theorem motivates checking for weakly constrained parameter directions, but neither proves that such directions exist in a particular network nor claims that optimization must follow them. Branch-level supervision, mutually exclusive hard routing, or explicit capability constraints break its assumptions.

3.3. Exact residual decomposition for nonlinear soft mixtures

The preceding theorem concerns non-uniqueness of the represented function. A separate question is whether a small physics residual of that function certifies the branch residuals. For the two-dimensional viscous Burgers operator, define

$$\mathcal{B}_\nu[v] = v_t + vDv - \nu\Delta v, \qquad D := \partial_x + \partial_y, \tag{4}$$

and write $u = \sum_{k=1}^{K} g_k u_k$, where $g_k \geq 0$ and $\sum_k g_k = 1$.

Theorem 2 (Nonlinear mixture-residual identity)**.** *Let $g_k, u_k \in C^{2,1}(\mathcal{Q})$, with two spatial derivatives and one time derivative continuous. Then*

$$\mathcal{B}_\nu[u] = \sum_{k=1}^{K} g_k \mathcal{B}_\nu[u_k] + \mathcal{I}_g + \mathcal{I}_{\mathrm{nl}}, \tag{5}$$

where

$$\mathcal{I}_g = \sum_k u_k(g_{k,t} - \nu\Delta g_k) - 2\nu \sum_k \nabla g_k \cdot \nabla u_k + u \sum_k u_k D g_k, \tag{6}$$

$$\mathcal{I}_{\mathrm{nl}} = u \sum_k g_k D u_k - \sum_k g_k u_k D u_k = -\operatorname{Cov}_g(u_k, Du_k). \tag{7}$$

Here $\operatorname{Cov}_g(a_k, b_k) = \sum_k g_k a_k b_k - (\sum_k g_k a_k)(\sum_k g_k b_k)$. Thus even exact knowledge of $\mathcal{B}_\nu[u]$ does not determine $\sum_k g_k \mathcal{B}_\nu[u_k]$: the weighted branch residual can be offset by gate derivatives and nonlinear cross-branch interaction. The specific form of this identity depends on the Burgers nonlinearity, but its core message—that the aggregate residual can be compensated by gate-derivative and nonlinear-covariance terms—holds for general nonlinear PDEs, with the covariance term replaced by the corresponding nonlinear-operator structure.

Proof. The product rule gives $u_t = \sum_k (g_k u_{k,t} + u_k g_{k,t})$, $Du = \sum_k (g_k D u_k + u_k D g_k)$, and $\Delta u = \sum_k (g_k \Delta u_k + 2\nabla g_k \cdot \nabla u_k + u_k \Delta g_k)$. Substitution into (4) separates the weighted branch operators from the gate-derivative terms. Adding and subtracting $\sum_k g_k u_k D u_k$ gives (7) and completes the identity. $\square$

The identity is structural rather than an optimization approximation. In particular, $\mathcal{I}_g$ is absent for a spatially and temporally constant gate, but $\mathcal{I}_{\mathrm{nl}}$ generally remains because the Burgers operator is nonlinear. Conversely, for a linear differential operator the nonlinear covariance vanishes, but product-rule terms generated by a nonconstant gate remain.

Proposition 2 (Zero mixture residual with unbounded branch residuals)**.** *Let $K = 2$, $g_1 = g_2 = 1/2$, and choose $\phi \in C_c^\infty(\mathcal{Q})$ such that $\phi D\phi \not\equiv 0$. For $M > 0$, set $u_1 = M\phi$ and $u_2 = -M\phi$. Then $u \equiv 0$ and hence $\mathcal{B}_\nu[u] \equiv 0$, while*

$$\|\mathcal{B}_\nu[u_k]\|_{L^2(\mathcal{Q})} \longrightarrow \infty, \qquad k = 1, 2, \tag{8}$$

as $M \to \infty$. Because ϕ is compactly supported in the interior, the construction preserves homogeneous initial and boundary data.

Proof. The mixture vanishes identically. With $L_\nu\phi = \phi_t - \nu\Delta\phi$,

$$\mathcal{B}_\nu[\pm M\phi] = \pm M L_\nu\phi + M^2\phi D\phi. \tag{9}$$

Dividing by M^2 gives convergence in L^2 to the nonzero function $\phi D\phi$; therefore both residual norms diverge quadratically. □

The converse direction requires additional information. Solving (5) for an active branch gives the following conditional certificate.

Proposition 3 (Conditional local branch certificate)**.** *At a point where $g_m \geq \gamma > 0$,*

$$|\mathcal{B}_\nu[u_m]| \leq \frac{1}{\gamma}\left(|\mathcal{B}_\nu[u]| + \sum_{k\neq m} g_k|\mathcal{B}_\nu[u_k]| + |\mathcal{I}_g| + |\mathcal{I}_{\mathrm{nl}}|\right). \tag{10}$$

Consequently, a small mixture residual certifies the active branch only when the active gate mass is bounded away from zero and the other-branch, gate-interaction, and nonlinear-interaction terms are also controlled.

Proof. Isolate $g_m\mathcal{B}_\nu[u_m]$ in (5), apply the triangle inequality, and use $g_m^{-1} \leq \gamma^{-1}$. □

Equations (5)–(10) distinguish two mechanisms that aggregate solution error alone conflates: output-level averaging can reduce prediction error as in (12), whereas operator-level compensation can reduce the PDE residual through $\mathcal{I}_g + \mathcal{I}_{\mathrm{nl}}$. We therefore treat the normalized magnitude

$$C_{\mathcal{B}} = 1 - \frac{\|\mathcal{B}_\nu[u]\|_2}{\|\sum_k g_k\mathcal{B}_\nu[u_k]\|_2 + \|\mathcal{I}_g\|_2 + \|\mathcal{I}_{\mathrm{nl}}\|_2 + \epsilon} \tag{11}$$

as a prospective residual-compensation diagnostic, not as a signed fraction or a causal estimator. Its empirical association with routing interventions must be tested rather than assumed.

3.4. Aggregation, capability gaps, and an error bound

For $\delta_k = u_k - u^\star$,

$$\left|\sum_k g_k\delta_k\right|^2 = \sum_k g_k|\delta_k|^2 - \frac{1}{2}\sum_{i,j} g_i g_j|\delta_i - \delta_j|^2. \tag{12}$$

The second term is nonnegative, so the mixture error may be lower than any single-branch error[20]. We define $G_{\text{agg}} = 1 - E_{\text{mix}}/(E_{\text{weighted}} + \epsilon)$ with $\epsilon = 10^{-12}$. It quantifies the gain of continuous aggregation relative to gate-weighted branch MSE. It does *not*, by itself, prove pointwise opposite-sign error cancellation; we therefore use "soft-aggregation dependence" rather than an "error-cancellation ratio." To test compensation directly, let $e_k(z) = u_k(z) - u^\star(z)$, $D = \mathbb{E}_z[\sum_k(g_k e_k)^2]$, and $X = \mathbb{E}_z[(\sum_k g_k e_k)^2] - D$. We report the cross-to-diagonal energy ratio $C_\times = X/(D+\epsilon)$. Negative $C_\times$ is direct evidence of cross-branch error cancellation, and $-C_\times$ is the fraction of diagonal weighted-error energy cancelled by the cross terms. The statistic is evaluated on the same held-out test grid as the other audit endpoints.

Let $\ell_k(z) = |u_k(z) - u^\star(z)|^2$, $k^\star = \arg\min_k \ell_k$, and $\Delta = \min_{j\neq k^\star}(\ell_j - \ell_{k^\star})$. Under local Lipschitz continuity, the oracle label is unchanged whenever the sum of the competing perturbation bounds is smaller than the corresponding capability gap. Freezing the experts is one sufficient way to hold this landscape fixed, but it is not unique; a two-time-scale optimizer may satisfy the same local condition.

Lemma 1 (Stability of the oracle label)**.** *Suppose each ℓ_k is uniformly L-Lipschitz in θ on a parameter neighborhood, i.e., $|\ell_k(\theta+\Delta\theta) - \ell_k(\theta)| \leq L\|\Delta\theta\|$. If at a point z the capability gap $\Delta(z) = \min_{j\neq k^\star}(\ell_j(z) - \ell_{k^\star}(z)) > 0$, then the oracle label $k^\star(z)$ is unchanged whenever $\|\Delta\theta\| < \Delta(z)/(2L)$.*

Proof. For any $j \neq k^\star$, $\ell_j(\theta+\Delta\theta) - \ell_{k^\star}(\theta+\Delta\theta) \geq \ell_j(\theta) - \ell_{k^\star}(\theta) - 2L\|\Delta\theta\| = \Delta(z) - 2L\|\Delta\theta\| > 0$, so $k^\star$ remains the pointwise minimizer. □

A moving-target interpretation is conditional. Assume that for fixed expert state Θ the gate objective is locally μ-strongly convex in Φ, has a unique differentiable minimizer $\Phi^\star(\Theta)$, and is followed by gradient flow. If $E(t) = \|\Phi(t) - \Phi^\star(\Theta(t))\|$ is differentiable and obeys $\dot{E} \leq -\mu E + C_\star\|\dot{\Theta}\|$, Grönwall's inequality gives

$$E(t) \leq e^{-\mu t}E(0) + C_\star\int_0^t e^{-\mu(t-s)}\|\dot{\Theta}(s)\|\,ds. \tag{13}$$

This is not a global convergence claim for a deep softmax gate.

Finally define $\varepsilon_{\mathrm{orc}}^2 = |\mathcal{Q}|^{-1} \int \min_k |u_k - u^\star|^2$, $\eta = |\mathcal{Q}|^{-1} \int (1 - g_{k^\star})$, and $\zeta = |\mathcal{Q}|^{-1} \int \sum_{j \neq k^\star} g_j |u_j - u^\star|^2$. If $|u_k - u^\star| \leq M$, Jensen's inequality yields

$$|\mathcal{Q}|^{-1} \|u_{\mathrm{mix}} - u^\star\|_2^2 \leq \varepsilon_{\mathrm{orc}}^2 + \zeta \leq \varepsilon_{\mathrm{orc}}^2 + M^2 \eta. \tag{14}$$

Equation (14) separates capability of the expert set from the cost of non-oracle routing.

3.5. From failure mechanism to a capability-first training paradigm

The analysis identifies a coherent failure loop. Persistently small soft weights attenuate the coupled expert gradient; weakly trained branches therefore remain weak. Because objectives depending on the mixture and gate statistics do not identify the branch decomposition, aggregate accuracy supplies no restoring force for those branches. For nonlinear soft mixtures, gate derivatives and gate-weighted covariance create additional compensation channels through which a small mixture residual can coexist with large branch residuals. Meanwhile, continuous expert updates move the pointwise capability ordering seen by the gate. Co-adaptive training thus asks the gate to learn a partition over branches whose competence is both unequal and changing.

These results support the *capability-before-routing principle*: the optimization process should establish a sufficiently stable capability ordering before routing competition is allowed to control branch-wise learning signal. This is a training paradigm because it specifies which functional object must be formed first, not merely how many epochs belong to each stage. In the notation of Eq. (14), capability formation reduces the oracle error $\varepsilon_{\mathrm{orc}}$; routing formation then reduces η and ζ without simultaneously redefining the oracle. The oracle-label stability lemma gives a local criterion for when this separation is meaningful.

CFST breaks the observed loop through the strongest form of this separation. Independent capability formation prevents early gate weights from suppressing branch updates. Freezing the branches during routing formation converts a moving capability terrain into a fixed learning target. A two-time-scale method, stop-gradient routing, or an explicit lower bound on branch capability could realize the same principle without literal freezing. CFST is therefore the controlled and experimentally validated implementation evaluated in this paper, while capability-before-routing is the general design rule.

Specialization-recovery theory establishes that mixtures can recover latent components under structured assumptions, suitable initialization, and favorable optimization [27]. CFST provides a numerical route to a favorable starting point in the present continuous-PDE setting: branch capability is constructed explicitly before specialization is delegated to the gate.

Remark 2. Finite-width networks need not represent every compactly supported perturbation in Theorem 1 exactly. The theorem nevertheless exposes the missing constraint that CFST restores at the algorithmic level: each complete branch is required to acquire capability before mixture compensation can hide its error.

The analysis shifts the question from whether the aggregate is accurate to how branch capability is formed. It identifies two testable objects: whether capability is established before routing competition, and whether a gate trained on a fixed capability terrain selects and preserves that capability. Section 4 turns these objects into the CFST protocol, matched controls, and a branch-reliability audit.

4. Branch-reliability audit and capability-first training

4.1. Problem setting, architecture, and evaluation reference

The main problem is the two-dimensional viscous Burgers equation

$$u_t + u(u_x + u_y) = \nu(u_{xx} + u_{yy}), \qquad \nu = 0.01/\pi, \tag{15}$$

on $[-1, 1]^2 \times [0, 1]$, with zero Dirichlet boundary values and

$$\begin{aligned} u(x, y, 0) = &- (1 - x^2)(1 - y^2)\{0.65 \sin[\pi(x + y)] - 0.35 \sin[\pi(x - y)] \\ &+ 0.20 \sin(2\pi x) \sin(\pi y)\}. \end{aligned} \tag{16}$$

The residual MoE contains a width-96, depth-5 tanh base; four heterogeneous residual experts (smooth: width 48/depth 3; iso-shock and directional-shock: width 96/depth 5; wave: width 64/depth 4); and a width-64, depth-3 local-convolutional gate with learned temperature $T_g = 0.8$ and 16 context neighbors, for 186,727 parameters. Cross-equation evidence uses an analytic KdV two-soliton solution and a two-dimensional Allen–Cahn phase field. A parameter-matched two-subnetwork APINN baseline has 186,429 parameters, a 0.16% difference from the main model.

The Burgers reference solves the conservative form with $f(u) = u^2/2$ in each spatial direction. A finite-difference WENO5 reconstruction is applied to a global Lax–Friedrichs flux split [37, 38]; diffusion uses a fourth-order centered interior stencil and a second-order stencil next to the boundary. Time integration uses SSP–RK3 [39], and the zero Dirichlet values are reimposed at every Runge–Kutta stage. With $h = \min(\Delta x, \Delta y)$, each step is bounded by

$$\Delta t = \min\left(0.35\, h/\max|u|, \frac{0.20}{\nu(\Delta x^{-2} + \Delta y^{-2})},\ \Delta t_{\text{out}}\right), \tag{17}$$

where the last term lands exactly on 41 equally spaced output times. The hierarchy uses 129^2, 257^2, and 513^2 spatial grids. Relative differences from 129 to 257 are 0.02210 globally and 0.04857 in the steep subset; from 257 to 513 they are 0.004658 and 0.01026, with maximum absolute difference 0.04974. The WENO5 teacher uses 21 output times and spatial indices $3, 7, \ldots, 511$ of the 513^2 solution. All main scalar metrics use the same times but spatial indices $1, 5, \ldots, 509$, giving 344,064 test points with exactly zero coordinate overlap with the teacher grid. Figure 4 alone uses the full 257^2 grid to obtain legible route maps.

4.2. Capability-first staged training and matched controls

CFST has two functional stages. The capability-formation stage updates every complete branch independently with its physics, initial-condition, boundary-condition, and reference losses. The routing-formation stage freezes all complete branches and fits the gate to their fixed pointwise capability terrain. The co-adaptive baseline receives the same reference information but propagates it through the mixture while updating experts and gate together. Table 2 gives both executable procedures.

Table 2: Capability-first staged training (CFST) and the co-adaptive baseline. All main endpoints are taken immediately after Step 2, before optional calibration.

Input	reference solution $u^{\star}$, collocation set $\mathcal{B}$, loss weights $(w_{\text{res}}, w_{\text{ic}}, w_{\text{bc}}) = (1, 5, 2)$
1	Initialize the model (1); train the shared base for 1,000 steps (lr 10^{-3})
2a	CFST: update each θ_k independently for 1,000 steps (lr 10^{-3}) using $\mathcal{L}_{\text{st},k}$; freeze all experts and train the gate for 1,000 steps (lr 5×10^{-4}) using $\mathcal{L}_{\text{gate}}$
2b	Co-adaptive: update all experts (lr 10^{-3}) and the gate (lr 5×10^{-4}) together for 1,000 steps using $\mathcal{L}_{\text{co}}$
3	Save the primary checkpoint and evaluate all metrics on the reference subset

The exact objectives are

$$\mathcal{L}_{\text{st},k} = \mathcal{L}_{\text{res},k} + 5\mathcal{L}_{\text{ic},k} + 2\mathcal{L}_{\text{bc},k} + 8\mathcal{L}_{\text{sup},k}, \tag{18}$$

$$\mathcal{L}_{\text{co}} = \mathcal{L}_{\text{phys}}(u_{\text{mix}}) + 8\mathcal{L}_{\text{sup,mix}} + \mathcal{L}_{\text{gate-match}}, \tag{19}$$

$$\mathcal{L}_{\text{gate}} = \mathbb{E}_z[(0.1 + c(z))D_{\text{KL}}(q(z)\|g(z))] + 0.01\mathcal{L}_{\text{bal}}, \tag{20}$$

where c is the gap between the two largest teacher probabilities. At each supervision point, absolute complete-branch reference errors are divided by their pointwise mean and converted to q by a softmax with sharpness $\beta = 8$, followed by a prespecified capability prior raised to 1.15 and renormalization. The CFST teacher remains fixed during routing formation; the co-adaptive teacher is refreshed every ten steps because its capability terrain continues to move. Each branch receives 1,800 supervision points per step; the co-adaptive union contains $4 \times 1{,}800$ points. Adam uses default moment coefficients, Xavier initialization, and unit-norm gradient clipping. An archived 750-step frozen-expert gate calibration was run as a sensitivity analysis and is reported separately from the primary endpoints.

To test whether the gain comes from branch-capability formation rather than a generic increase in training effort, we match supervision information, initialization, update counts, and computational budget. All experiments were run on a laptop with an NVIDIA GeForce RTX 4060 Laptop GPU (8 GB memory), using the CUDA backend of PyTorch. Across the ten formal seeds, a single CFST/co-adaptive training run took 281.7 ± 2.8 s and 328.6 ± 3.5 s, about 102 minutes in total for all twenty runs. Peak allocated GPU memory was 0.62/1.81 GB, respectively. The twenty Allen–Cahn control runs (two modes times ten seeds) took about 21 minutes in total. Individual APINN and KdV gate-introduction runs were substantially cheaper and are not listed separately. All re-evaluations were performed on the 344,064-point subset of the 513^2 conservative WENO5 reference, with a single model evaluation taking on the order of ten seconds.

A lower-budget 2×2 screening experiment crosses physics-only/reference-guided information with blocked/interleaved updates, using ten paired seeds per cell. This control determines whether the benefit comes merely from blocking updates or from forming capability with branch-level information; each cell uses 270 base steps and 300 main updates. The primary high-budget comparison uses seeds 42–51, 1,000 base steps, and 1,000 updates in each capability/routing stage. Both primary protocols expose the model to $4 \times 1800 \times 1000 = 7.2 \times 10^6$ supervised point-events before the checkpoint.

The comparison matches the numerical reference, paired initialization, base steps, updates per expert, and supervised point-event ledger. The intended intervention is the supervision topology: CFST assigns reference-supported capability losses to complete branches before routing, whereas co-adaptation assigns mixture-level supervision during routing competition. Across the ten formal runs, training times are 281.7 ± 2.8 s and 328.6 ± 3.5 s, and peak allocated GPU memory is 0.62 and 1.81 GB for CFST and co-adaptation, respectively.

4.3. Branch-reliability audit and statistical analysis

The audit separates four logically distinct qualification questions. *Aggregate validity* asks whether the soft mixture approximates the reference solution. *Branch capability* asks whether complete branches are globally and locally usable, independent of their current routing mass. *Gate–capability alignment* asks whether accurate branches exist and whether the gate selects them. *Routing robustness* asks whether accuracy survives hardening, flattening, or branch removal. Utilization and training-trajectory diagnostics are retained as mechanistic evidence, but they are not treated as certificates of numerical capability. This layered structure prevents a favorable aggregate error or a balanced load from answering a stronger question than it measures.

All relative L_2 errors are dimensionless. We report mixture error; complete-branch global error; responsibility-region absolute RMSE; oracle error; soft-routing regret; effective experts $\exp[-\sum_k \bar{g}_k \log \bar{g}_k]$; and soft-aggregation gain. The logged coupled-gradient diagnostic is

$G_k^{(s)} = \|\nabla_{\theta_k}\mathcal{L}_{\text{joint}}^{(s)}\|_2$, measured after backpropagation and before gradient clipping and the Adam update. Because $\mathcal{L}_{\text{joint}}$ is evaluated through u_{mix}, this quantity includes gate-induced scaling. It is not the gradient of an independently evaluated branch loss.

Capability regions are defined before training from derivatives of the numerical reference, not from the learned gate or from post-training branch errors. Let $r_\nabla = |\nabla u^\star|/(Q_{0.9}(|\nabla u^\star|)+\epsilon)$, $r_\Delta = |u^\star_{xx}+u^\star_{yy}|/(Q_{0.9}(|u^\star_{xx}+u^\star_{yy}|)+\epsilon)$, $r_{xy} = |u^\star_{xy}|/(Q_{0.9}(|u^\star_{xy}|)+\epsilon)$, and $\tau = t/T$. The four clipped scores are

$$s_{\text{sm}} = [1.1 - r_\nabla]_0^{1.5}, \tag{21}$$

$$s_{\text{iso}} = [r_\nabla(1.2 - 0.7[r_{xy}]_0^{1.5})]_0^{1.5}, \tag{22}$$

$$s_{\text{dir}} = [r_\nabla[1.1 r_{xy}]_0^{1.5}(0.35 + 0.65\tau)]_0^{1.5}, \tag{23}$$

$$s_{\text{wave}} = [(r_\Delta - 0.35 r_\nabla)(0.25 + 0.75\tau)]_0^{1.5}, \tag{24}$$

where $[a]_0^{1.5} = \min(1.5, \max(0, a))$. Each grid point is assigned to the largest score. This construction makes the region audit reproducible while keeping it independent of the trained router.

For a capability region C_r, the reported branch error is

$$E^{\text{cap}}_{r,k} = \frac{[\sum_{z\in C_r} |u_k(z) - u^\star(z)|^2]^{1/2}}{\max\{[\sum_{z\in C_r} |u^\star(z)|^2]^{1/2}, 10^{-10}\}}. \tag{25}$$

Every prespecified Burgers region is nonempty. Route entropy is $H_{\text{route}} = -|\mathcal{E}|^{-1}\sum_z\sum_k g_k(z)\log[g_k(z)+10^{-12}]$, and minimum load is $\min_k \bar{g}_k$. Counterfactuals use the primary checkpoint and the same 513^2 evaluation subset. Top-1 replaces g by its argmax one-hot vector. Post-hoc temperature uses $\widetilde{g}_k(\tau) = g_k^{1/\tau}/\sum_j g_j^{1/\tau}$; $\tau = 2$ flattens the learned distribution without retraining. Highest-load deletion sets the branch with largest $\bar{g}_k$ to zero and renormalizes the remaining weights.

Table 3: Branch-reliability audit: qualification question, primary diagnostic, and failure interpretation. No single metric certifies all properties.

Question	Primary diagnostic	Failure indicated
Is the aggregate accurate?	Soft-mixture relative L_2	Poor aggregate approximation
Can each complete branch stand alone?	Global branch L_2 and worst branch	Hidden globally unusable branches
Does a branch work where intended?	Ex-ante capability-region error	Specialty failure not explained by global extrapolation
Is the expert set capable?	Oracle L_2	No branch is locally accurate
Does the gate select capability?	Routing regret and responsibility RMSE	Accurate branches exist but are not selected
Does accuracy require continuous averaging?	G_{agg}; Top-1, $\tau = 2$, deletion ratios	Sensitivity to routing hardening or removal
Is utilization healthy?	Soft load, effective experts, route entropy	Starvation or concentration; not a certificate of function quality

For the ten paired Burgers seeds, effects are defined as co-adaptive minus CFST. We report sample mean ± sample standard deviation, deterministic paired-bootstrap 95% confidence intervals from 200,000 resamples (random seed 20260809) [40], and exact two-sided sign-flip tests over all 2^{10} sign assignments. Holm adjustment is applied across the nine endpoints in Table 5 [41]. Mixture and worst-branch errors are the two primary endpoints; the remaining tests quantify utilization and routing robustness.

The training mechanism is thereby mapped to testable endpoints: CFST and co-adaptation are matched in information, budget, and initialization, while the audit separately measures aggregate performance, branch capability, gate alignment, and routing robustness. Section 5 follows this evidence chain by isolating the role of capability information and then examining primary performance, training trajectories, routing interventions, and cross-equation transfer.

5. Results

5.1. Information and update chronology in a 2 × 2 control

Table 4: Information source × update chronology (ten seeds; mean ± sample standard deviation).

Configuration	Mixture relative L_2	Worst branch L_2	Effective experts
P–B: physics, blocked	0.37467 ± 0.03494	0.37496 ± 0.03491	3.698
P–I: physics, interleaved	0.37468 ± 0.03495	0.37496 ± 0.03491	3.798
R–B: reference, blocked	0.35580 ± 0.02763	0.39318 ± 0.02141	3.239
R–I: reference, interleaved	0.35351 ± 0.02795	0.39318 ± 0.02141	3.462

Reference guidance reduces mixture error by 0.0189 under blocked updates and by 0.0212 under interleaved updates, with 19 of the 20 paired contrasts in the same direction, whereas the mean blocked–interleaved difference is small. The low-budget screen does not show an improvement in worst-branch error from reference guidance alone; it instead rules out simple regrouping of otherwise identical updates as the explanation for the complete CFST treatment. The main comparison therefore evaluates the combined intervention in which branch-level information forms capability before gate-mediated competition. Figure 2 visualizes the information and chronology effects without treating the screen as a standalone branch-reliability certificate.

5.2. CFST restores branch reliability while retaining accurate aggregation

Table 5: Primary pre-calibration results on the conservative WENO5 reference (ten paired seeds). Differences are co-adaptive minus CFST; p is the exact two-sided sign-flip value.

Metric	CFST	Co-adaptive	Paired difference [95% CI]	p_{exact}
Soft-mixture relative L_2	0.2108 ± 0.0252	0.1600 ± 0.0379	-0.0508 $[-0.0636, -0.0310]$	0.00391
Worst complete-branch L_2	0.3336 ± 0.0426	21.51 ± 8.34	21.177 $[16.437, 26.200]$	0.00195
Oracle L_2	0.1319 ± 0.0118	0.1124 ± 0.0178	-0.0195 $[-0.0269, -0.0115]$	0.00391
Soft-aggregation gain	0.1484 ± 0.0265	0.9520 ± 0.0266	0.8036 $[0.7796, 0.8258]$	0.00195
Soft-routing regret	0.0021 ± 0.0006	0.0430 ± 0.0262	0.0409 $[0.0261, 0.0564]$	0.00195
Effective experts	3.550 ± 0.273	2.664 ± 0.156	-0.887 $[-1.052, -0.716]$	0.00195
Top-1 / soft error	1.058 ± 0.022	1.383 ± 0.109	0.325 $[0.255, 0.398]$	0.00195
$\tau = 2$ / soft error	0.987 ± 0.017	2.899 ± 1.001	1.912 $[1.384, 2.555]$	0.00195
Delete highest-load / soft error	1.090 ± 0.088	1.520 ± 0.385	0.430 $[0.213, 0.695]$	0.00195

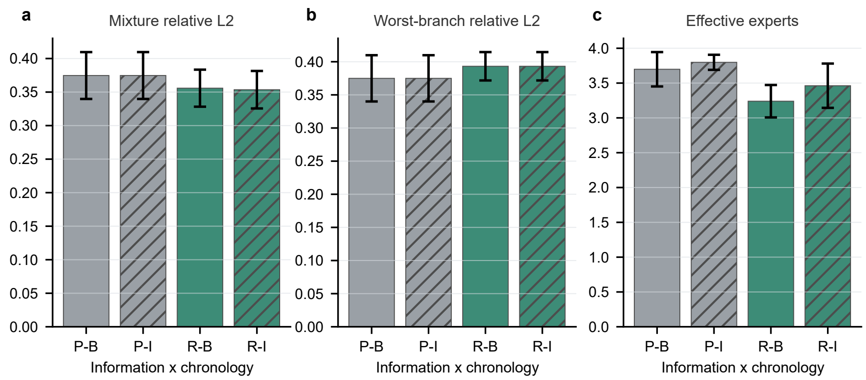

Figure 2: Grouped bars for the 2×2 factorial experiment (10 paired seeds; mean $\pm$ sample SD). (a) Mixture relative L_2; (b) worst complete-branch relative L_2; (c) effective experts. Gray denotes physics-only information and green denotes reference guidance; unhatched and hatched bars denote blocked and interleaved updates, respectively. Information source affects mixture error more strongly than update chronology in this low-budget screen, while worst-branch error is not improved by reference guidance alone.

CFST reaches a substantially improved aggregate accuracy relative to the earlier low-budget protocol while decisively improving the internal solver. Co-adaptation is lower in soft-mixture error by 0.0508 (95% CI $[-0.0636, -0.0310]$), so aggregate accuracy alone favors it. The qualification audit changes the conclusion: the worst complete-branch difference is positive on every seed, with mean 21.18 and 95% CI $[16.44, 26.20]$ (exact $p = 0.00195$, Holm-adjusted $p = 0.0176$). Its worst branch is approximately 64.5 times the CFST value of 0.3336 ± 0.0426. The capability-region audit in Fig. 3 locates the defect inside and outside the intended specialization region, excluding local-expert extrapolation as the explanation. Figure 4 completes the spatial picture: CFST maintains a structured four-branch partition, whereas co-adaptive routing concentrates and leaves the directional branch inaccurate over large parts of the domain. The result exposes a measurable aggregate-accuracy–branch-reliability trade-off that the proposed audit makes visible.

The oracle and routing endpoints localize the source of this trade-off. Even under pointwise best-branch selection, the CFST oracle error is 0.1319 ± 0.0118, compared with 0.1124 ± 0.0178 for co-adaptation; hence part of the aggregate gap is already present in the trained branch set rather than being introduced by the gate. Conversely, CFST has much smaller soft-routing regret (0.0021 ± 0.0006 versus 0.0430 ± 0.0262), which rules out poor gate fitting as the principal cause of its higher mixture error. The exact decomposition in Eq. (26) supplies direct evidence for the remaining mechanism. The cross-to-diagonal energy ratio is negative for every co-adaptive seed, with mean -0.172 ± 0.101, so cross-branch terms cancel $17.2\% \pm 10.1\%$ of the diagonal weighted-error energy. For CFST the ratio is positive on every seed (0.303 ± 0.062), and no cancellation is observed. Joint optimization therefore lowers aggregate error partly by arranging compensating branch errors, whereas branchwise qualification followed by freezing restricts that freedom. The lower co-adaptive mixture error is consequently coupled to an internally fragile decomposition rather than to a uniformly more capable expert system.

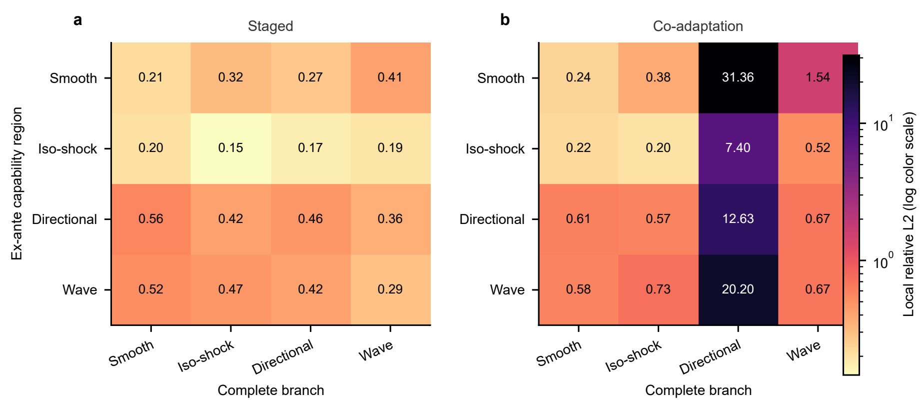

Figure 3: Capability-region error matrix on the conservative WENO5 reference. Rows denote ex-ante capability regions and columns denote complete branches. Each cell reports the mean local relative L_2 over 10 paired seeds; colors use a logarithmic scale and numbers show the untransformed values. (a) CFST keeps every cell between 0.146 and 0.561. (b) Co-adaptation raises the directional-shock branch to 7.40–31.36 across regions. Regions are computed from reference derivatives before training and are independent of the learned gate.

5.2.1. Training trajectories reveal the starvation loop

Figures 6 and 7 use the lower-budget 300-step diagnostic protocol, which preserves the same supervision topology as the primary comparison and logs the feedback loop at finer relative intervals. Routing mass progressively concentrates on the smooth and iso-shock branches while the directional-shock

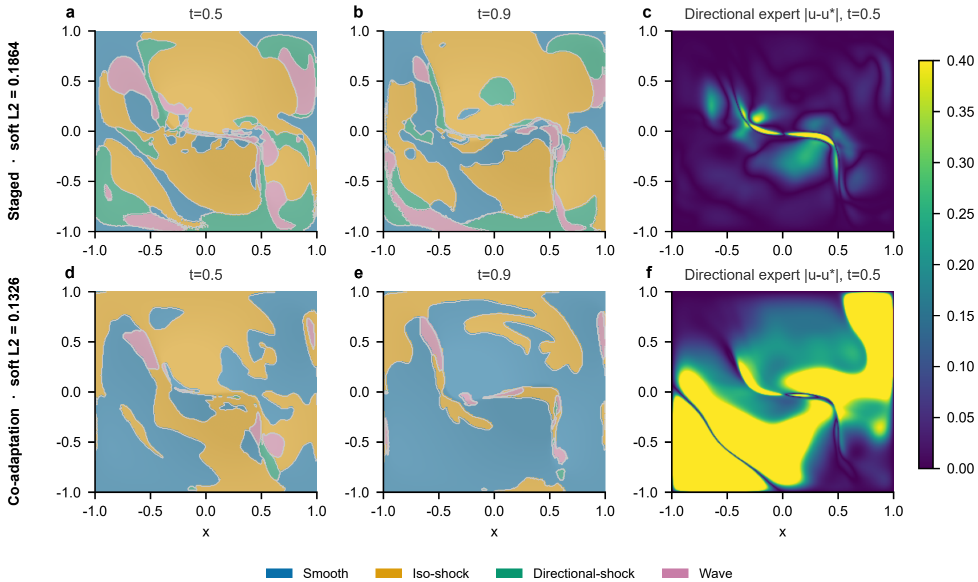

Figure 4: Routing partitions and directional-branch errors on the full 257^2 WENO5 visualization grid (seed 43, primary pre-calibration checkpoints). (a–c) CFST and (d–f) co-adaptive. Panels (a,b,d,e) show argmax routing at $t = 0.5$ and $t = 0.9$, overlaid on the reference field; colors identify Smooth, Iso-shock, Directional-shock, and Wave experts, and light contours mark route boundaries. Panels (c,f) show $|u_{\mathrm{dir}} - u^{\star}|$ at $t = 0.5$. Their common Viridis scale is capped at 0.4 to preserve contrast for the CFST panel. Values above 0.4 are clipped only for display; the reported scalar errors use the separate 344,064-point subset of the 513^2 reference.

mean weight falls to 0.014; its gate-weighted parameter-gradient signal falls with it. This trajectory connects the analytical gate scaling to the observed loss of directional-branch capability.

Figure 7 tracks the complementary part of the loop. CFST freezes capability losses and oracle labels during routing formation, whereas co-adaptation continues to change both the median capability gap and the pointwise best-branch labels. The early oracle-label flip rate reaches approximately 0.66, forcing the co-adaptive gate to learn against a rapidly moving target.

5.2.2. Routing-robustness interventions

CFST converts branch health into functional routing robustness. Switching to Top-1 routing, flattening the gate with $\tau = 2$, and deleting the highest-

Figure 5: Paired-seed results on the conservative WENO5 reference (seeds 42–51, primary checkpoints). (a) Soft-mixture relative L_2; co-adaptation is lower in 9 of 10 pairs, while the pale band marks seed 42 where it is higher. (b) Worst complete-branch relative L_2 on a logarithmic axis; co-adaptation is more than one order of magnitude higher for every paired seed. Lines connect outcomes across seeds for each protocol and do not represent temporal trajectories.

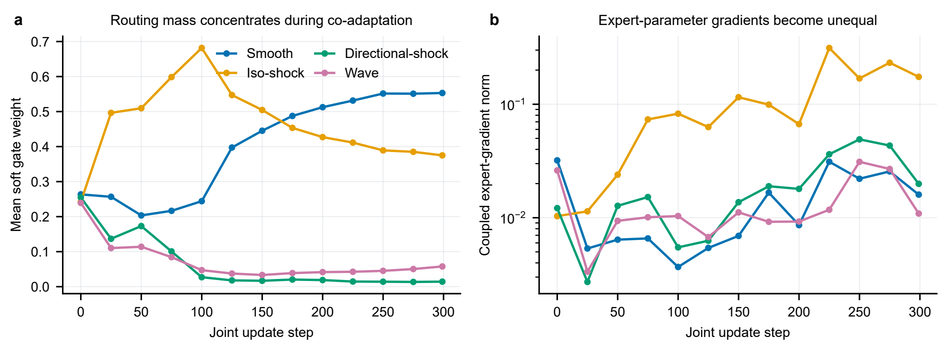

Figure 6: Gate-weight concentration and coupled expert gradients in the lower-budget mechanistic trace (seed 42; 300 co-adaptive updates; fixed 20,000-point WENO5 evaluation subset). (a) Domain-mean soft gate weight for the four experts. (b) Coupled expert-parameter gradient norm $G_k^{(s)} = \|\nabla_{\theta_k} \mathcal{L}_{\text{joint}}^{(s)}\|_2$, recorded after backpropagation and before gradient clipping and the Adam update, on a logarithmic axis. The diagnostic is the realized gate-weighted update signal under the mixture objective.

load branch give CFST error ratios of 1.058 ± 0.022, 0.987 ± 0.017, and 1.090 ± 0.088, respectively. The co-adaptive ratios rise to 1.383 ± 0.109,

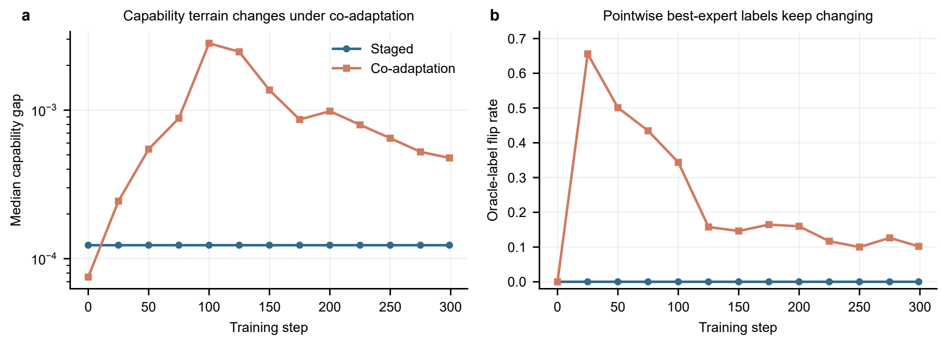

Figure 7: Capability-terrain trajectory in the lower-budget mechanistic trace (seed 42; fixed 20,000-point WENO5 evaluation subset). (a) Median pointwise capability gap $\Delta(z)$ on a logarithmic axis. (b) Fraction of evaluation points whose oracle label changes relative to the previous logged checkpoint. Co-adaptation produces a non-monotone capability terrain and an early flip rate of approximately 0.66, whereas both quantities remain fixed under CFST. The first checkpoint has no previous checkpoint and is plotted with flip rate zero by convention.

2.899 ± 1.001, and 1.520 ± 0.385. All paired intervals in Table 5 exclude zero. Thus CFST retains useful aggregate accuracy when the routing regime is altered, whereas the co-adaptive solution depends materially on soft aggregation.

5.3. Increasing capability formation progressively improves the decomposition

To test the dose–response relation between capability formation and branch reliability, we re-evaluated five Burgers training schedules ($f = 0\%, 25\%, 50\%, 75\%, 100\%$, three seeds) on the conservative WENO5 reference. Here f denotes 0, 75, 150, 225, or 300 independent updates per branch before the gate is introduced. For $f < 100\%$, a cold gate is introduced and branches and gate are then updated jointly for 750 steps. At $f = 100\%$, the branches receive all 300 independent updates, the gate receives 180 gate-only updates, and the subsequent 750 steps update only the gate. Figure 9 shows that the mixture L_2 falls from 0.53 at $f = 0\%$ to 0.36 at $f = 100\%$; the worst-expert L_2 drops from about 5.2 to 0.40, more than an order of magnitude; effective experts rise from 2.5 to 3.6 and the minimum load from 0.02 to 0.13. Across the sweep, more capability formation produces a healthier decomposition: the worst-branch error decreases by more than an order of magnitude, effective branch use increases, and the minimum load rises. The maximum absolute error is non-monotone and is

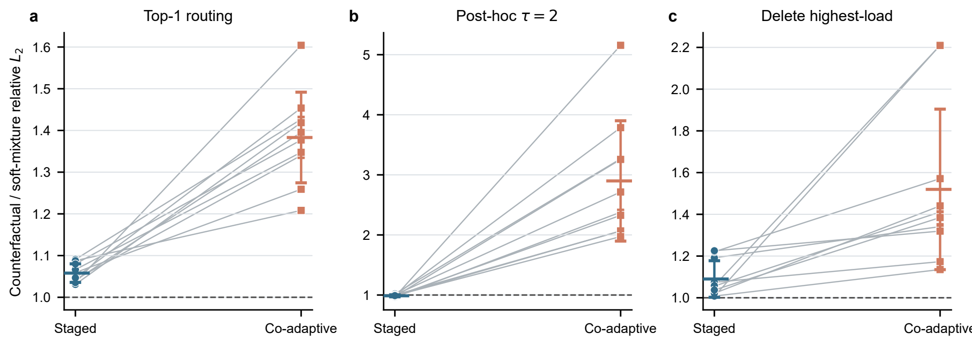

Figure 8: Routing counterfactuals at the primary checkpoint on the same 344,064-point subset of the 513^2 WENO5 reference (10 paired seeds). Each line joins the same seed across protocols; filled markers and bars show mean $\pm$ sample SD. The ordinate is counterfactual relative L_2 divided by the original soft-mixture error, and the dashed line marks one. Top-1 uses argmax routing; $\tau = 2$ applies $\widetilde{g}_k \propto g_k^{1/2}$; "Delete highest-load" removes the branch with the largest mean soft load and renormalizes the rest.

largest at $f = 100\%$, showing that reliability should be assessed jointly with pointwise accuracy rather than substituted for it.

5.4. Supporting cross-equation and cross-architecture evidence

For Allen–Cahn, joint refinement from 201 to 401 gives a 0.145% global difference and a 0.674% interface difference. The reference describes a moving circular phase boundary. We define the positive-phase interior by $u^\star > 0.5$, the negative-phase exterior by $u^\star < -0.5$, and the diffuse interface by $|u^\star| \leq 0.5$; these reference-defined regions are independent of the learned gate. CFST/co-adaptive mixture errors are $0.036 \pm 0.010/0.487 \pm 0.320$. The interface branch's within-specialty error is $0.220 \pm 0.079/3.45 \pm 1.36$, while its co-adaptive load falls to 0.002 ± 0.007. The moving-interface problem therefore reproduces both starvation and the CFST recovery observed for Burgers[28]. Figure 11 summarizes the interface metrics.

APINN controls are re-evaluated on the conservative WENO5 reference. The 186,429-parameter two-subnetwork version gives a soft-mixture error of 0.320 ± 0.113, a worst-subnetwork error of 0.881 ± 0.077, and a Top-1/soft ratio of 1.985 ± 0.639. The official-size two-subnetwork version gives 0.250 ± 0.069, 2.601 ± 0.696, and 3.101 ± 0.683. The four-subnetwork parameter-matched version gives 0.390 ± 0.043, 1.454 ± 0.129, and 1.016 ± 0.024. All

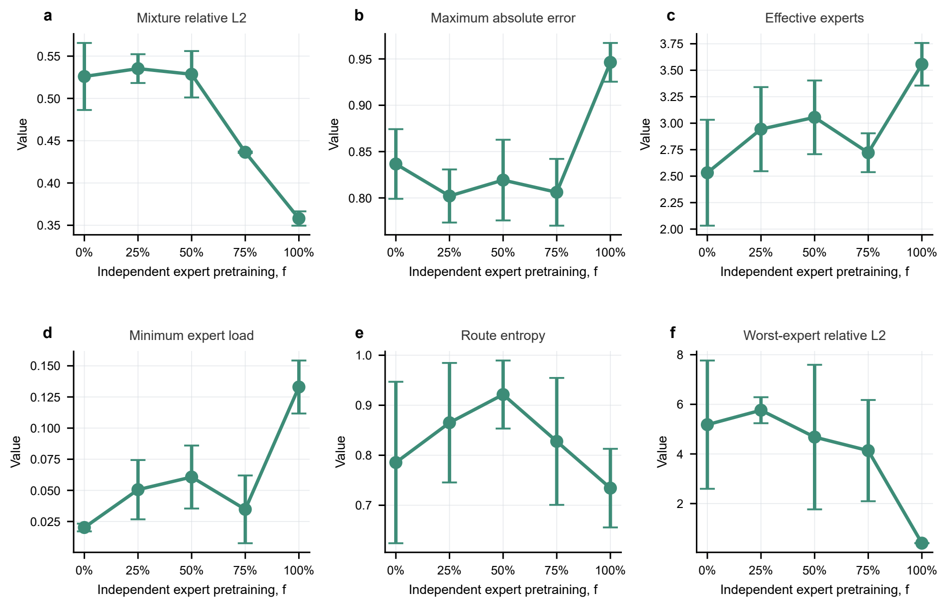

Figure 9: Exploratory Burgers gate-introduction schedule sweep (conservative WENO5 reference, three seeds; mean $\pm$ sample SD). (a) Mixture relative L_2; (b) maximum absolute error; (c) effective experts; (d) minimum expert load; (e) route entropy; (f) worst-expert L_2. The horizontal axis is the fraction of the 300 independent expert updates completed before gate introduction. Because the $f < 100\%$ schedules subsequently update experts and gate jointly whereas $f = 100\%$ uses gate-only updates, the curves describe complete training schedules rather than an isolated timing intervention.

three versions exhibit the same aggregate–branch separation. We retain the APINN ratios as supporting cross-architecture evidence and do not rank absolute performance. Figure 12 reports the aggregate, worst-subnetwork, and Top-1/soft metrics for all three versions.

The KdV schedule sweep independently reproduces the capability-before-routing effect against an analytic two-soliton solution. Here $f = 0\%, 25\%, 50\%, 75\%, 100\%$ corresponds to 0, 220, 440, 660, or 880 independent updates per branch before a common 2,200-step joint fine-tuning phase; $f = 100\%$ additionally receives 400 gate-only steps. Figure 13 visualizes the routing partition in the x–t plane. At $f = 100\%$, the three branches form a structured partition along the soliton band and their full-domain loads are near-uniform (0.35/0.33/0.32). At $f = 0\%$, routing concentrates approximately 79% of the domain on the shock branch and the

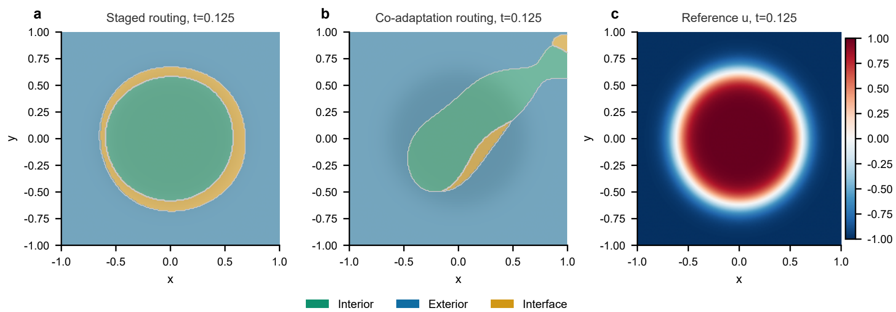

Figure 10: Allen–Cahn routing at $t = 0.125$ for seed 42. (a) CFST and (b) co-adaptive argmax routing; colors identify the interior, exterior, and interface branches. (c) Reference phase field, whose zero-level neighborhood marks the moving circular interface. CFST assigns a visible band to the interface branch, whereas co-adaptation nearly removes that band. The region error and load statistics in Fig. 11 are computed over the full evaluation grid.

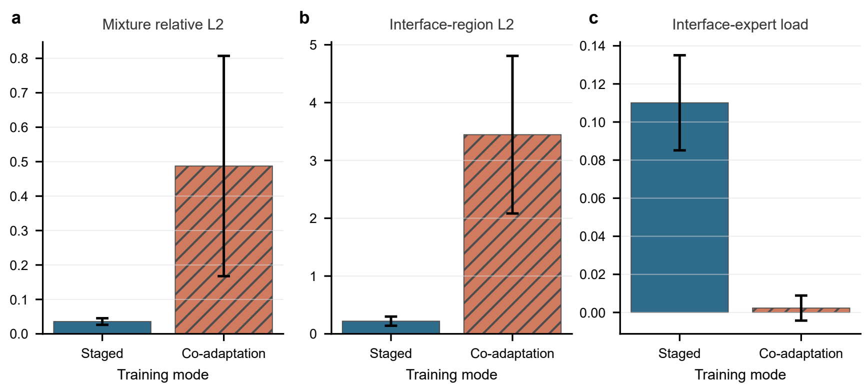

Figure 11: Allen–Cahn interface metrics (10 paired seeds; mean $\pm$ sample SD). Blue unhatched bars denote CFST and coral hatched bars co-adaptation. (a) Mixture relative L_2 over the full domain; (b) interface-branch relative L_2 in $|u^\star| \leq 0.5$; (c) mean soft load of the interface branch. Co-adaptation is worse in (a,b) and assigns almost no routing mass to the interface branch in (c).

mixture L_2 degrades from 0.025 to 0.30. As f rises from 0% to 25%, the mixture L_2 drops from 0.32 to 0.045 and the worst-branch error from 3.9 to about 0.8. At $f = 100\%$, route entropy reaches 0.09 and the minimum load rises to 0.31. The Burgers, Allen–Cahn, and KdV results therefore align across nonlinear transport, moving interfaces, and dispersive waves.

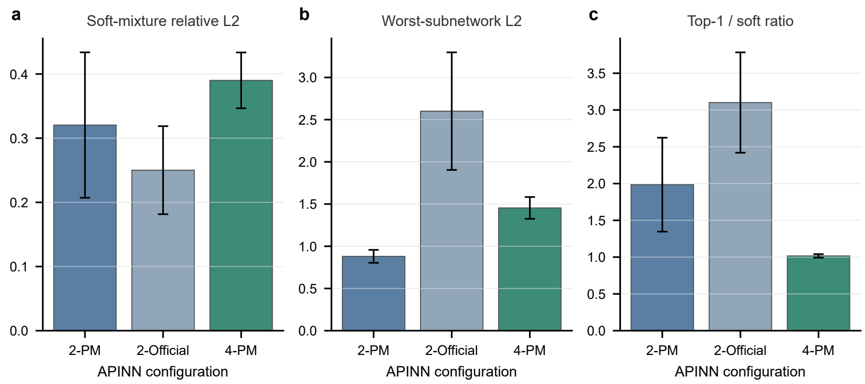

Figure 12: APINN control metrics on the conservative WENO5 reference (three seeds per configuration; mean ± sample SD). The configurations are a parameter-matched two-subnetwork model (2-PM), the official-size two-subnetwork model (2-Official), and a parameter-matched four-subnetwork model (4-PM). (a) Soft-mixture relative L_2; (b) worst complete-subnetwork relative L_2 over the full domain; (c) Top-1/soft error ratio. The comparison is used to test transfer of the aggregate–branch separation across architectures, not to rank their absolute accuracy.

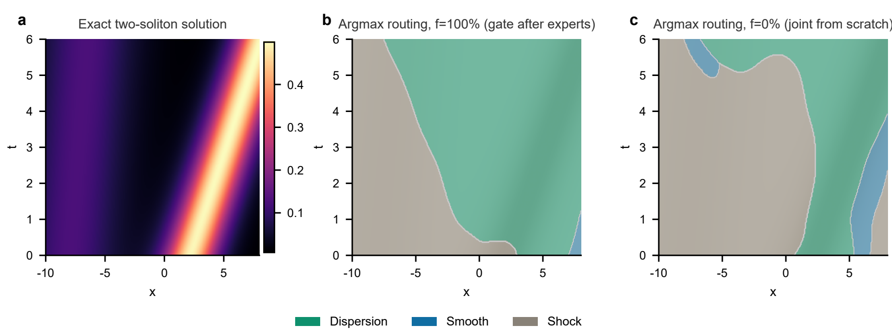

Figure 13: KdV two-soliton solution and x–t routing partitions (seed 42; interaction window $x \in [-10, 8]$). (a) Exact solution. (b) $f = 100\%$, where the dispersion, smooth, and shock experts form a structured partition along the soliton trajectories. (c) $f = 0\%$, where argmax routing concentrates on the shock expert. Colors identify expert identity and are shared across panels (b,c); the exact-solution color scale in (a) represents amplitude rather than routing. The reported loads and errors use the full test domain.

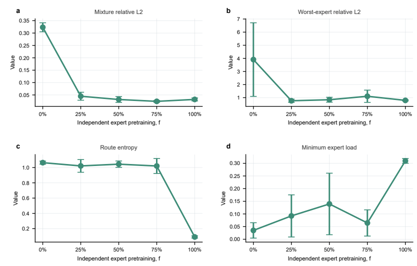

Figure 14: KdV gate-introduction schedule sweep (three seeds; mean $\pm$ sample SD). (a) Mixture relative L_2; (b) worst complete-branch relative L_2; (c) route entropy; (d) minimum mean branch load. The horizontal axis is the fraction of 880 independent branch updates completed before gate introduction. All schedules receive 2,200 joint steps, while $f = 100\%$ additionally receives 400 gate-only steps.

Together, these results form a progression from local mechanism to external validity. The paired control rules out simple chronology as the explanation, the Burgers study shows that CFST breaks the starvation loop and restores branch capability, routing interventions verify functional robustness, and the Allen–Cahn, KdV, and APINN controls support transfer of the aggregate–branch distinction. The next section discusses the implications, use cases, and extensions of this training paradigm.

6. Discussion

6.1. From a qualification gap to a training paradigm

This work turns the aggregate–branch qualification gap into an auditable, interpretable, and actionable training problem. The audit distinguishes the mixture output from the learned expert division, the analysis identifies a self-reinforcing failure loop created by gate-suppressed learning and mixture compensation, and CFST breaks that loop by forming capability before learning routing.

CFST changes the optimization geometry rather than the model capacity. During capability formation, every complete branch receives a direct learning signal that is not attenuated by the gate; during routing formation, the gate sees fixed capability gaps instead of a target that changes with every branch update. The non-identifiability theorem and nonlinear residual identity provide the theoretical basis for this separation: an aggregate objective admits compensating decompositions and does not force branch residuals to be small. The high-budget protocol reaches a soft-mixture error of 0.2108 ± 0.0252, reduces the worst-branch error by approximately 64.5-fold relative to co-adaptation, and is markedly more stable under all three routing interventions. The lower co-adaptive mixture error reveals a genuine accuracy–reliability trade-off rather than invalidating the treatment: CFST removes the hidden dependence on compensating unreliable branches. Allen–Cahn and KdV reproduce the capability-before-routing advantage, while AP-INN controls show that the aggregate–branch gap extends beyond the main architecture. CFST therefore represents a paradigm in which capability formation precedes routing competition, rather than the only implementation of that paradigm; methods that reduce gate-induced gradient suppression, stabilize the capability terrain, or explicitly maintain branch-capability lower bounds may improve reliability through the same mechanism.

This trade-off follows directly from the geometry of mixture error. With branch errors $e_k = u_k - u^\star$ and weighted errors $\widetilde{e}_k = g_k e_k$,

$$\left\| \sum_k \widetilde{e}_k \right\|_2^2 = \sum_k \|\widetilde{e}_k\|_2^2 + 2 \sum_{i<j} \langle \widetilde{e}_i, \widetilde{e}_j \rangle. \tag{26}$$

Co-adaptation optimizes the left-hand side directly and can reduce it through negative cross terms even when individual branch errors remain large. This occurs in all ten co-adaptive runs: the mean cross-to-diagonal energy ratio is -0.172 ± 0.101. CFST instead reduces complete-branch losses before routing and then freezes the branch set; its corresponding ratio is positive in all ten runs (0.303 ± 0.062), so it does not obtain aggregate accuracy through cross-branch error cancellation. The oracle gap confirms that the resulting restriction partly resides in the learned branch ensemble, while the much smaller routing regret confirms that it is not primarily a gate-estimation failure. Equation (26) therefore connects the observed accuracy gap to the same compensatory non-identifiability that motivates branch-level qualification.

When an MoE–PINN must support hard routing, expert pruning, gate-temperature adjustment, or subsystem reuse, complete-branch capability directly determines whether the model remains modular. CFST forms independently usable branches at low computational cost and makes the solution less dependent on delicate error compensation by the original soft gate. The resulting workflow is to establish aggregate validity, audit complete-branch capability and gate alignment, intervene on routing, and apply capability-first training when the decomposition does not pass the audit.

6.2. Limitations and future work

To isolate the effects of training order, supervision topology, and gate–expert coupling, the main experiments use the same numerical or analytic reference information for CFST and co-adaptation and match paired initialization, update budget, and evaluation grids. The reference solution therefore serves as a controlled variable and a common capability scale, rather than a logical prerequisite of the capability-before-routing principle. The Burgers residual analysis and routing interventions, together with the Allen–Cahn, KdV, and APINN controls, support the mechanism from complementary directions. The ten-seed paired validation currently centers on Burgers and Allen–Cahn, while the KdV schedule sweep and APINN variants use three seeds.

Future work proceeds along three directions. First, we will extend capability formation and routing learning to settings without reference solutions, using physics-based capability proxies such as multiscale space–time integral weak-form consistency to test whether conservation and weak-residual constraints alone can form qualified branches, and to characterize the effective boundary of such proxies. Second, we will combine the capability-before-routing principle with dynamic expert management, exploring capability representations updated under independent physical-qualification gates and controlled expert generation conditioned on both regional novelty and insufficient qualification of existing experts, with the goal of producing genuinely effective new experts when existing experts fail while vetoing meaningless architecture growth when they remain qualified. Third, we will run the necessary positive controls and generalize CFST's freeze-then-train geometry to alternative mechanisms such as two-time-scale optimization and stop-gradient routing, in order to probe the boundary of the capability-before-routing principle under different optimization geometries.

7. Conclusions

Aggregate accuracy is not a certificate of expert decomposition in soft MoE–PINNs. This paper turns that distinction into a closed qualification framework: audit aggregate validity, complete-branch capability, gate–capability alignment, and routing-intervention robustness; explain failure through gate–expert coupling and mixture non-identifiability; and repair it through the capability-before-routing principle. CFST is a tested, effective, and low-cost realization of this paradigm. It gives every complete branch the opportunity to become a competent numerical solver and then learns routing over a fixed capability terrain. On Burgers, the high-budget CFST reaches a soft-mixture relative L_2 error of 0.2108 ± 0.0252, lowers the worst-branch error by approximately 64.5-fold relative to co-adaptation, and remains substantially more stable under Top-1 routing, gate flattening, and highest-load deletion. Allen–Cahn and KdV reproduce the same advantage. Whenever branches may later be selected, removed, or rerouted, the decomposition must be qualified separately from the aggregate, and capability should be formed before routing competition controls learning.

CRediT authorship contribution statement

Xiang Li: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Liping Huang:** Supervision, Project administration, Funding acquisition, Writing – review & editing.

Data and code availability

The source code, configurations, numerical reference solutions, processed result tables, per-seed metrics, and figure source data supporting this study are publicly available at https://github.com/username-a/Capability-First-Training-for-Reliable-MoE-PINNs. Large neural-network checkpoints are not included because they can be regenerated from the deposited configurations and scripts.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, manuscript organization, and LaTeX preparation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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